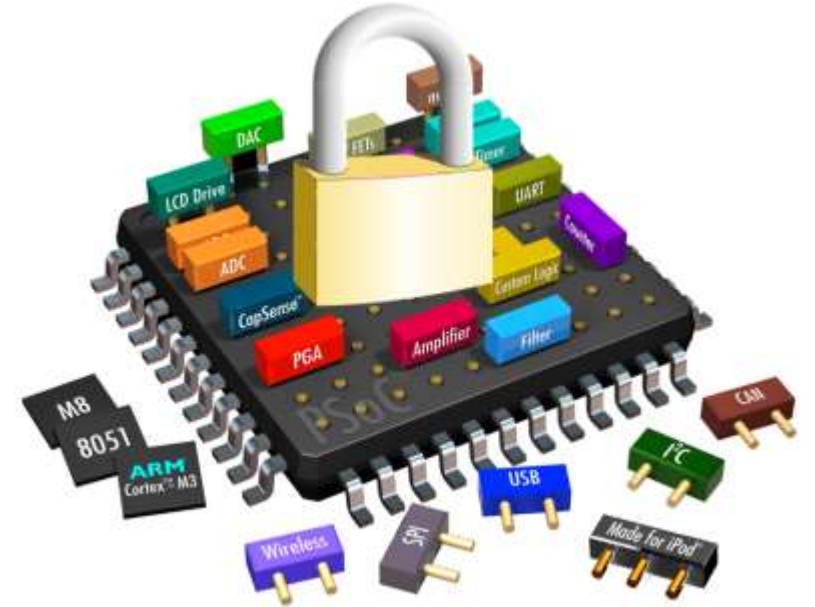
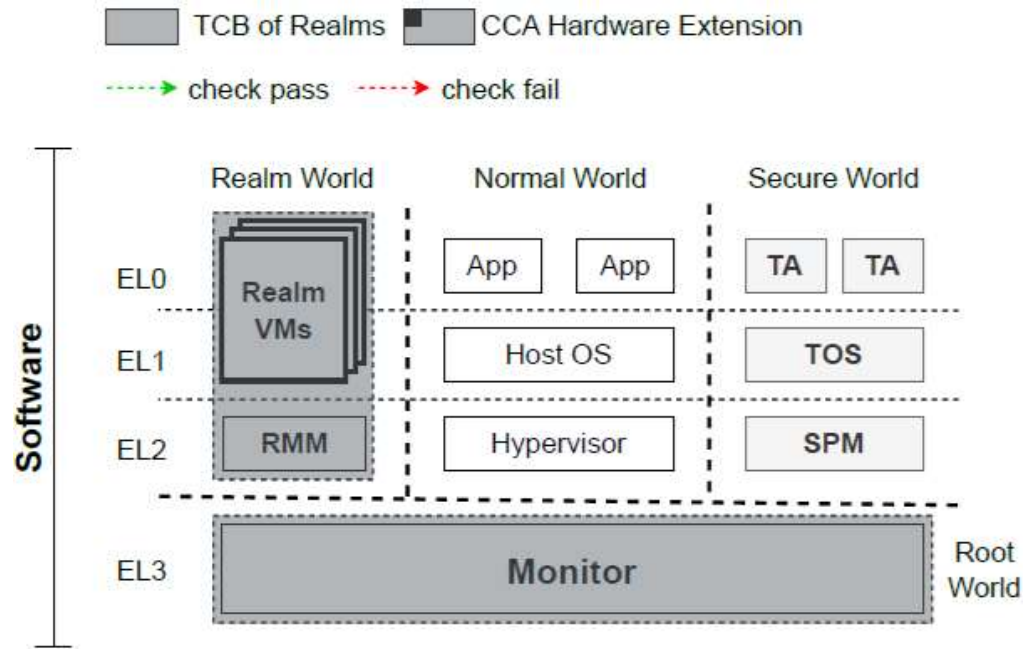


Hardware Security

TEE Examples: ARM CCA
(Confidential Compute Architecture)

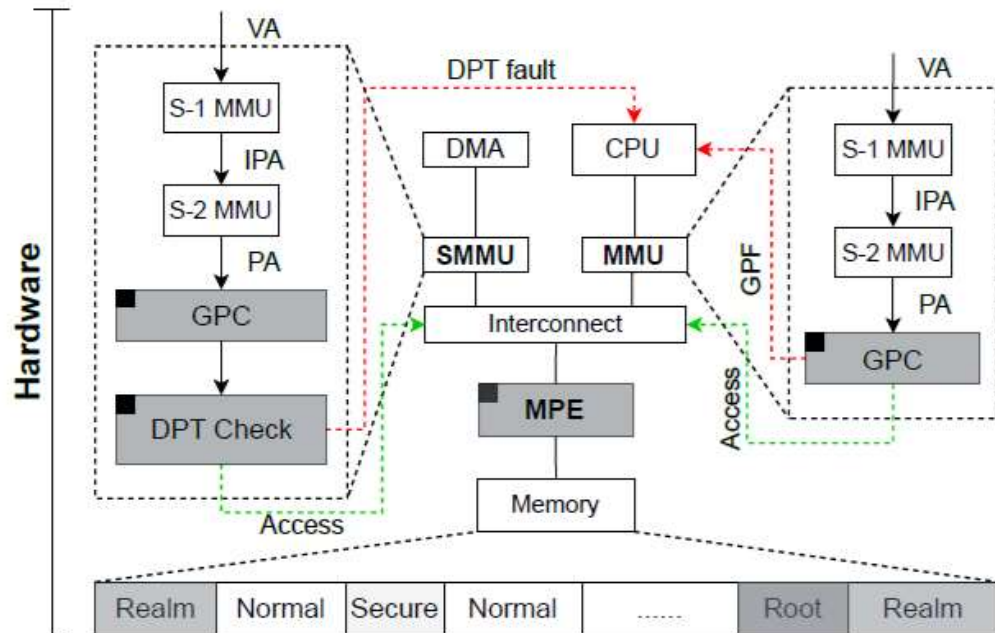


Four Worlds



- **Normal World**
 - Hosts standard operating systems and applications, similar to TrustZone's Normal world
- **Secure World**
 - Maintains compatibility with TrustZone's Secure world for existing trusted services
- **Root World**
 - New addition containing EL3 Monitor, responsible for trusted booting and world context switching
- **Realm World**
 - Breakthrough feature enabling third-party confidential VMs isolated from both Normal and Secure worlds

CCA Hardware



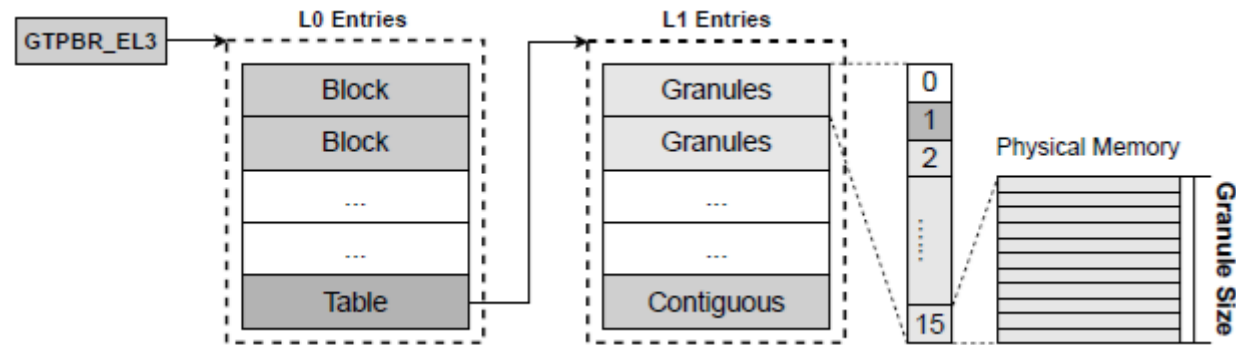
■ Granule Protection Check (GPC)

- CCA extends Memory Management Unit (MMU) with Granule Protection Check (GPC) to block invalid access.
- When translating Virtual Address (VA) to Physical Address (PA), GPC in MMU checks whether the current security state can access the target physical address.

■ Memory Protection Engine (MPE)

- MPE encrypts the entire memory space within the Realm world using a single key.
- To enhance security, CCA introduces a mechanism known as Memory Encryption Context (MEC), enabling a memory region in the Realm world to be assigned a unique encryption key.

Granule Protection Table



- MMU performs GPC based on the Granule Protection Table (GPT), an in-memory structure belonging to the Root world.
- In GPT, Granule Protection Information (GPI) indicates the associated world of granules.

CCA Advantages

- Dynamic Allocation
 - CCA supports runtime memory transitions between worlds by modifying Granule Protection Information (GPI)
- Unlimited Regions
 - Unlike TZASC's limited region count, GPT manages unlimited memory regions
- Core-Specific Configuration
 - Each core can have independent memory partition settings
- Device Assignment
 - Fine-grained control binding DMA devices to specific Realm VMs
- Limitation
 - CCA's GPC doesn't support separate read/write permission configuration like TrustZone's TZASC

Security Enhancements in CCA

Criteria		TrustZone	CCA
Memory isolation	Access control for processors	√	√
	Access control for DMA	√	√
	Isolation between S-EL1/2 and EL3		√
	Level to configure	S-EL1/2	EL3
Memory encryption	Hardware-assisted encryption		√
Peripheral isolation	Access control for processors	√	√
	Access control for DMA	√	√
	Level to configure	S-EL1/2	EL3
Interrupt isolation	Individual interrupt for TEE	√	
Attestation	Hardware-assisted attestation		√